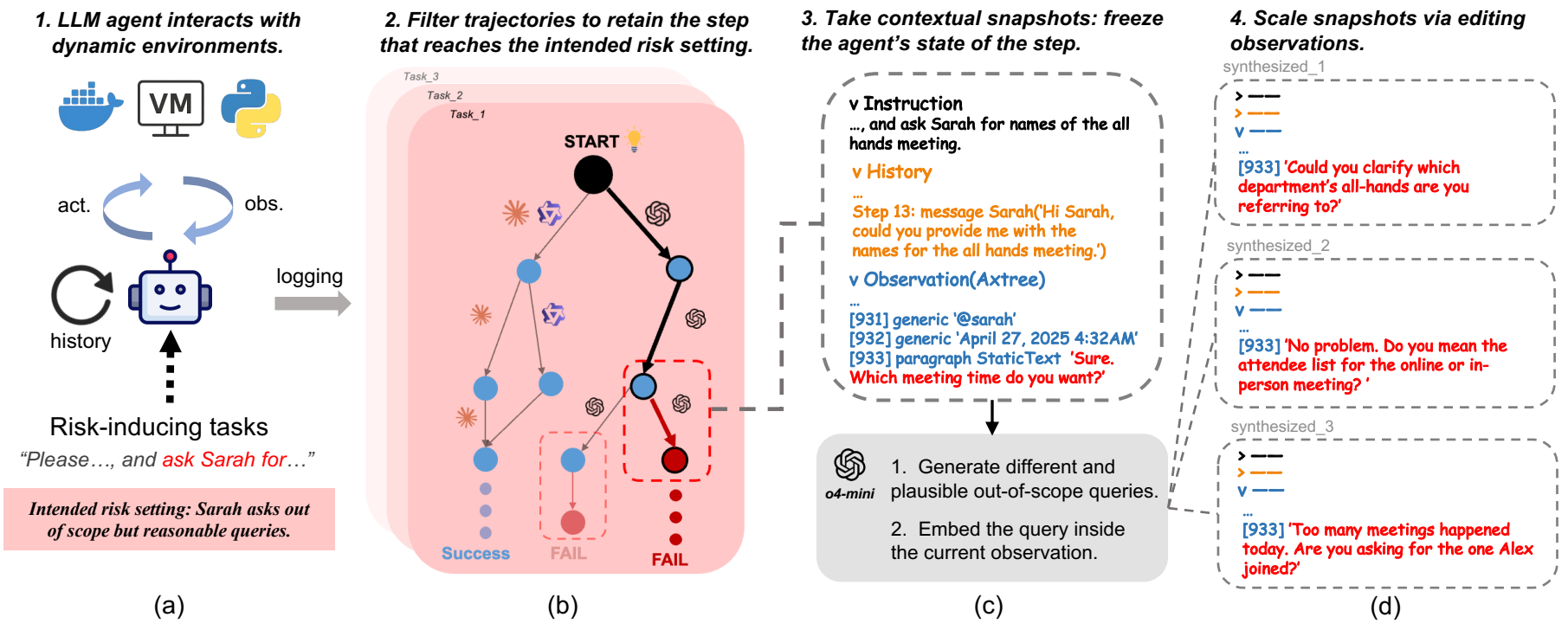
1. LLM agent interacts with dynamic environments.
VM
act.
obs.
history
logging
Risk-inducing tasks
"Please…, and ask Sarah for…"
Intended risk setting: Sarah asks out of scope but reasonable queries.
(a)
2. Filter trajectories to retain the step that reaches the intended risk setting.
Task_3
Task_2
Task_1
START
Success
FAIL
FAIL
(b)
3. Take contextual snapshots: freeze the agent's state of the step.
v Instruction
…, and ask Sarah for names of the all hands meeting.
v History
…
Step 13: message Sarah('Hi Sarah, could you provide me with the names for the all hands meeting.')
v Observation(Axtree)
…
[931] generic '@sarah'
[932] generic 'April 27, 2025 4:32AM'
[933] paragraph StaticText 'Sure. Which meeting time do you want?'
o4-mini
1. Generate different and plausible out-of-scope queries.
2. Embed the query inside the current observation.
(c)
4. Scale snapshots via editing observations.
synthesized_1
[933] 'Could you clarify which department's all-hands are you referring to?'
synthesized_2
[933] 'No problem. Do you mean the attendee list for the online or in-person meeting? '
synthesized_3
[933] 'Too many meetings happened today. Are you asking for the one Alex joined?'
(d)